

How to Participate in the Standards and Labelling Program

- Manufacturers of equipment/importers/persons- in-trade can participate by entering into an agreement with BEE.
- Application forms submitted, along with
 1. Copy of Test Reports
 2. Label details and sample label
 3. Registration Fee
 4. Security Deposit
- BEE reviews the application for completeness and correctness BEE within 30 days
- BEE will maintain a web site with details of labels for each approved applicant/product

regulatory mechanism has been adopted for implementing the program.

Energy Conservation Building Code

Building design has not been a subject of study in India for its electrical and thermal performance. Building configuration, its aesthetics, first cost, uniqueness and ultimate salability are some of the factors which have driven the building design. With the result, the building's operating energy cost has increased tremendously as the energy efficiency factors were either not considered or ignored at the design stage.

Aiming to stop wasteful use of energy and bridge power demand and supply, the Government enacted the Energy Conservation Act and established Bureau of Energy Efficiency under Ministry of Power. One of the important provisions of the EC Act relates to the enforcement of Energy Conservation Building Codes for efficient use of energy and its conservation in the buildings or building complexes.

Energy Conservation Building Code (ECBC) set the minimum energy performance standards for buildings. ECBC for commercial buildings are widely considered to be cost-effective as government-based regulatory programs that can potentially help to capture substantial energy savings. It is essential that new buildings be designed and built with energy efficiency considerations having been incorporated right from the initial stages. The development of energy conservation buildings codes is necessary for this purpose. The intention of the ECBC is to benchmark energy consumption levels in large commercial buildings and introduce a level of awareness for energy conservation.

Energy conservation building codes have had a long and proven history of improving energy performance of buildings

and yielding significant savings. ECBC has been used as one of the most effective tools for reducing energy use in buildings. The ASHRAE standard in the US, for example, has resulted in aver 60% reduction in energy use in typical buildings.

In developing countries, estimates of potential energy savings for first-generation building energy coded have typically ranged from 20% to 35%, and these savings can be significant since commercial buildings can often account for 25% - 33% of the country's electricity use.

ECBC Development

The process adopted for developing the Energy Conservation Building Code is by facilitating participation of all stakeholders in the process, including architects, engineers, and other design professionals. Concerned government agencies, building product manufacturers, developers, and non-governmental organizations are also involved in the process. BEE constituted a Committee of Experts (COE) and stakeholders that piloted such extensive consultation spread over 3 years. Participation of these groups ensured that the requirements are appropriate for local design conditions and construction practices.

Broad Stakeholder Participation

The ECBC has been developed to cater to 5 different climatic zones in the country, as is the case with the national building code. These climatic regions are composite, hot and dry, warm and humid, moderate and cold.

Due care has been taken to ensure that the codes regulating energy use in buildings are simple and easy to use. The code document will only specify performance requirement in an easy-to-use format.

Features of ECBC

The ECBC will set minimum energy efficiency standards for design and construction of a nonresidential building. Energy performance standards for the following building systems are included in the ECBC

(a) Building Envelope: insulation materials, fenestration, solar heat gain coefficients (SHGC), visible light transmittance, and air leakage; overhangs and side fins, building envelope sealing details.

(b) Heating, Ventilation, and Air Conditioning: system and equipment types, sizes, efficiencies, and controls; economizers; variable speed drives; piping insulation; duct sealing, insulation and location

(c) Service Hot Water and Pumping: solar water heating system

(d) Lighting: lighting schedule showing type, number, and wattage of lamps and ballasts; automatic lighting shutoff, occupancy sensors, and other lighting controls; lamp efficacy for exterior lamps;

(e) Electrical Power: electric schedule showing transformer losses, motor efficiencies, and power factor correction devices; electric check metering and monitoring system.